```
import pandas as pd
```

```
import numpy as np
```

```
df = pd.read_excel("/content/Dataset.xlsx")
df.head()
```

	Rank	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sales
0	1	Wii Sports	Wii	2006	Sports	Nintendo	41.49	29.02
1	2	Super Mario Bros.	NES	1985	Platform	Nintendo	NaN	3.58
2	3	Mario Kart Wii	Wii	2008	Racing	Nintendo	15.85	NaN
3	4	Wii Sports Resort	Wii	2009	Sports	Nintendo	15.75	11.01

Next steps: Generate code with df | New interactive sheet

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19 entries, 0 to 18
Data columns (total 11 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Rank          19 non-null     int64
 1   Name          19 non-null     object
 2   Platform      19 non-null     object
 3   Year          19 non-null     int64
 4   Genre         19 non-null     object
 5   Publisher     19 non-null     object
 6   NA_Sales      19 non-null     float64
 7   EU_Sales      19 non-null     float64
 8   JP_Sales      19 non-null     float64
 9   Other_Sales   19 non-null     float64
 10  Global_Sales  19 non-null     float64
dtypes: float64(5), int64(2), object(4)
memory usage: 1.8+ KB
```

```
df.shape
```

```
(19, 11)
```

```
df.describe()
```

	Rank	Year	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Gl
count	19.000000	19.000000	19.000000	19.000000	19.000000	19.000000	
mean	10.000000	2002.105263	14.302895	7.618421	3.760789	2.874737	
std	5.627314	8.924963	8.285388	6.011926	2.590003	2.563722	
min	1.000000	1984.000000	7.010000	0.400000	0.240000	0.470000	
25%	5.500000	1997.500000	9.260000	4.345000	2.230000	1.730000	
50%	10.000000	2006.000000	11.325000	7.570000	3.685000	2.260000	
75%	14.500000	2008.500000	15.360000	9.045000	4.460000	2.930000	
max	19.000000	2013.000000	41.490000	29.020000	10.220000	10.570000	

```
df.columns
```

```
Index(['Rank', 'Name', 'Platform', 'Year', 'Genre', 'Publisher', 'NA_Sales',
       'EU_Sales', 'JP_Sales', 'Other_Sales', 'Global_Sales'],
      dtype='object')
```

```
df.isnull().sum()
```

	0
Rank	0
Name	0
Platform	0
Year	0
Genre	0
Publisher	0
NA_Sales	3
EU_Sales	2
JP_Sales	1
Other_Sales	2
Global_Sales	2

dtype: int64

```
df.dtypes
```

	0
Rank	int64
Name	object
Platform	object
Year	int64
Genre	object
Publisher	object
NA_Sales	float64
EU_Sales	float64
JP_Sales	float64
Other_Sales	float64
Global_Sales	float64

dtype: object

```
df[df.select_dtypes(include="number").columns] = df.select_dtypes(include="n
```

```
sales_columns = ["NA_Sales", "EU_Sales", "JP_Sales", "Other_Sales", "Global_
```

```
df[sales_columns] = df[sales_columns].fillna(df[sales_columns].median())
```

```
print(df.isnull().sum())
```

```
Rank            0
Name            0
Platform        0
Year            0
Genre           0
Publisher       0
NA_Sales        0
EU_Sales        0
JP_Sales        0
Other_Sales     0
Global_Sales    0
dtype: int64
```

```
import matplotlib.pyplot as plt
import seaborn as sns
```

Global Sales of selected games.

```
selected_games = df.head(6)

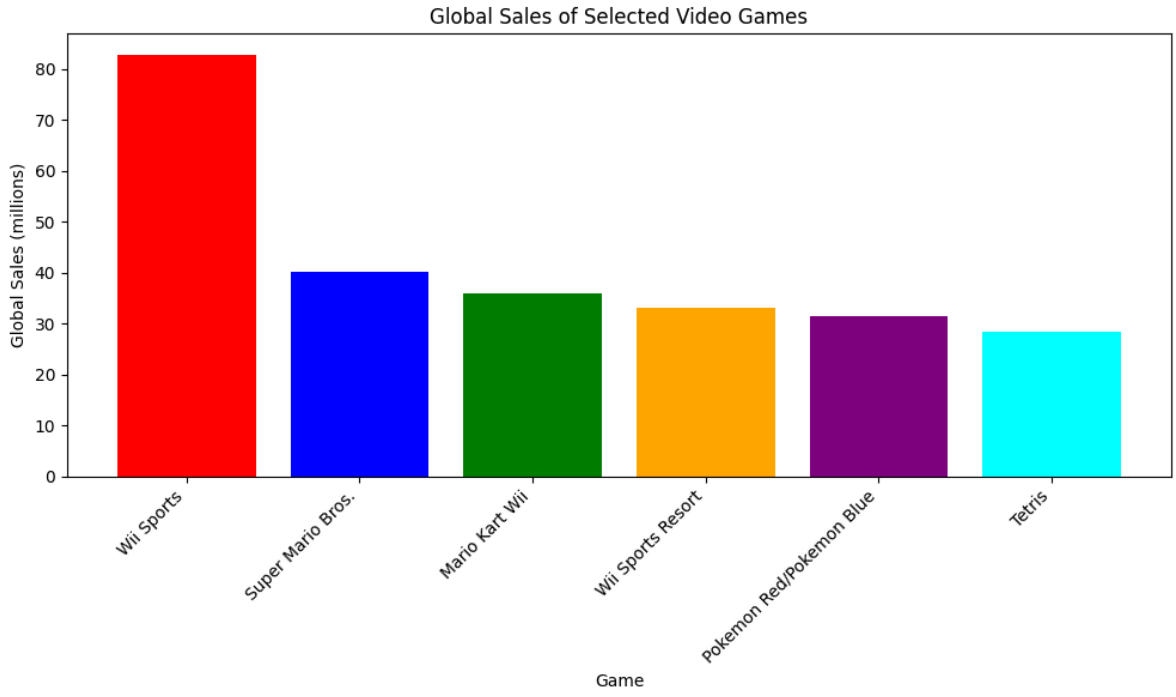
plt.figure(figsize=(10, 6))

plt.bar(
    selected_games["Name"],
    selected_games["Global_Sales"],
    color=["red", "blue", "green", "orange", "purple", "cyan"]
)

plt.title("Global Sales of Selected Video Games")
plt.xlabel("Game")
plt.ylabel("Global Sales (millions)")

plt.xticks(rotation=45, ha="right")

plt.tight_layout()
plt.show()
```



Average global sales across years

```
year_sales = df.groupby("Year")["Global_Sales"].mean()

plt.figure(figsize=(10, 6))

plt.plot(
    year_sales.index,
    year_sales.values,
    marker="D",
    linestyle="-",
    linewidth=2,
    markersize=7
)

plt.title("Average Global Game Sales Across Years")
plt.xlabel("Year")
plt.ylabel("Average Global Sales (millions)")

plt.grid(True, linestyle="--", alpha=0.5)

plt.tight_layout()
plt.show()
```



North American Sales and Global **Sales**

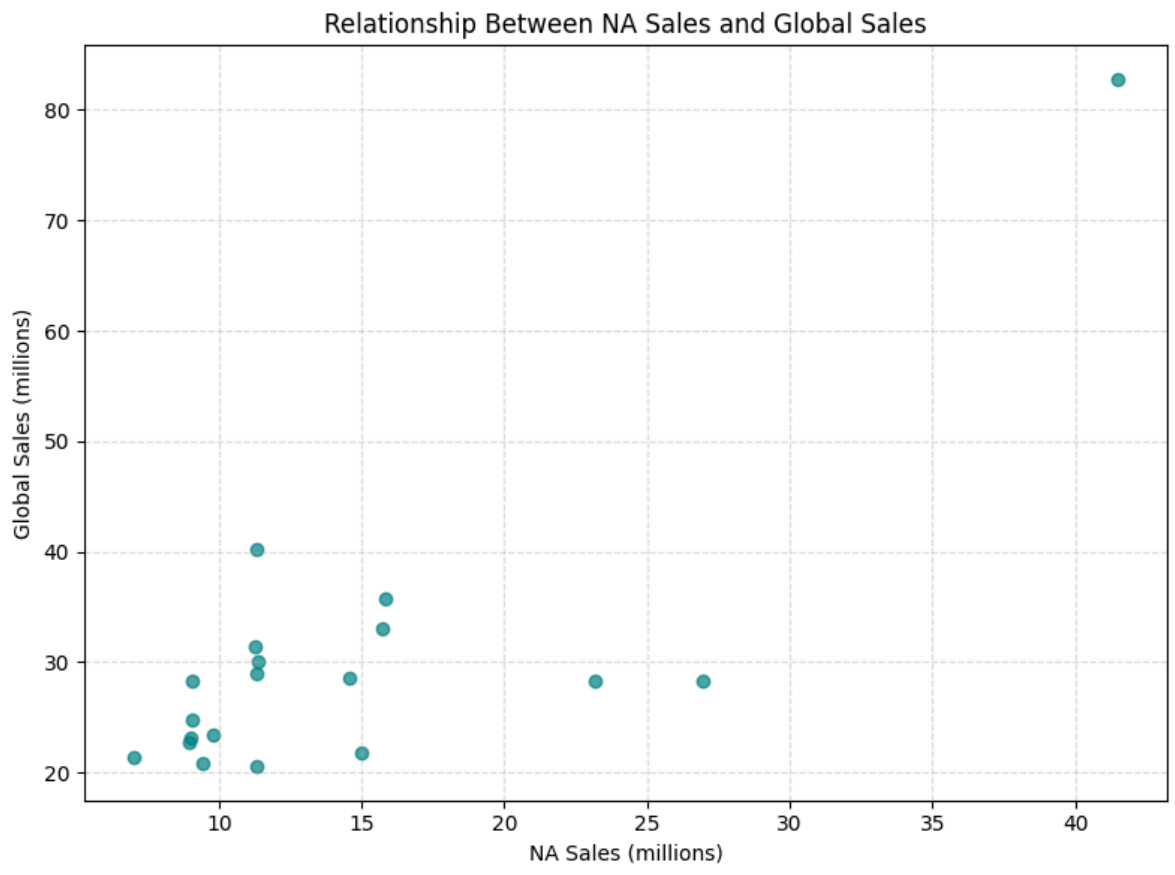
```
plt.figure(figsize=(8, 6))

plt.scatter(
    df["NA_Sales"],
    df["Global_Sales"],
    color="teal",
    alpha=0.7
)

plt.title("Relationship Between NA Sales and Global Sales")
plt.xlabel("NA Sales (millions)")
plt.ylabel("Global Sales (millions)")

plt.grid(True, linestyle="--", alpha=0.4)

plt.tight_layout()
plt.show()
```



NA Sales and Global Sales

```
selected_games = df.head(6)

x = range(len(selected_games))

plt.figure(figsize=(10, 6))

plt.bar(
    [i - 0.2 for i in x],
    selected_games["NA_Sales"],
    width=0.4,
    label="NA Sales",
    color="steelblue"
)

plt.plot(
    x,
    selected_games["Global_Sales"],
    marker="D",
    linewidth=2,
    label="Global Sales",
    color="darkorange"
)

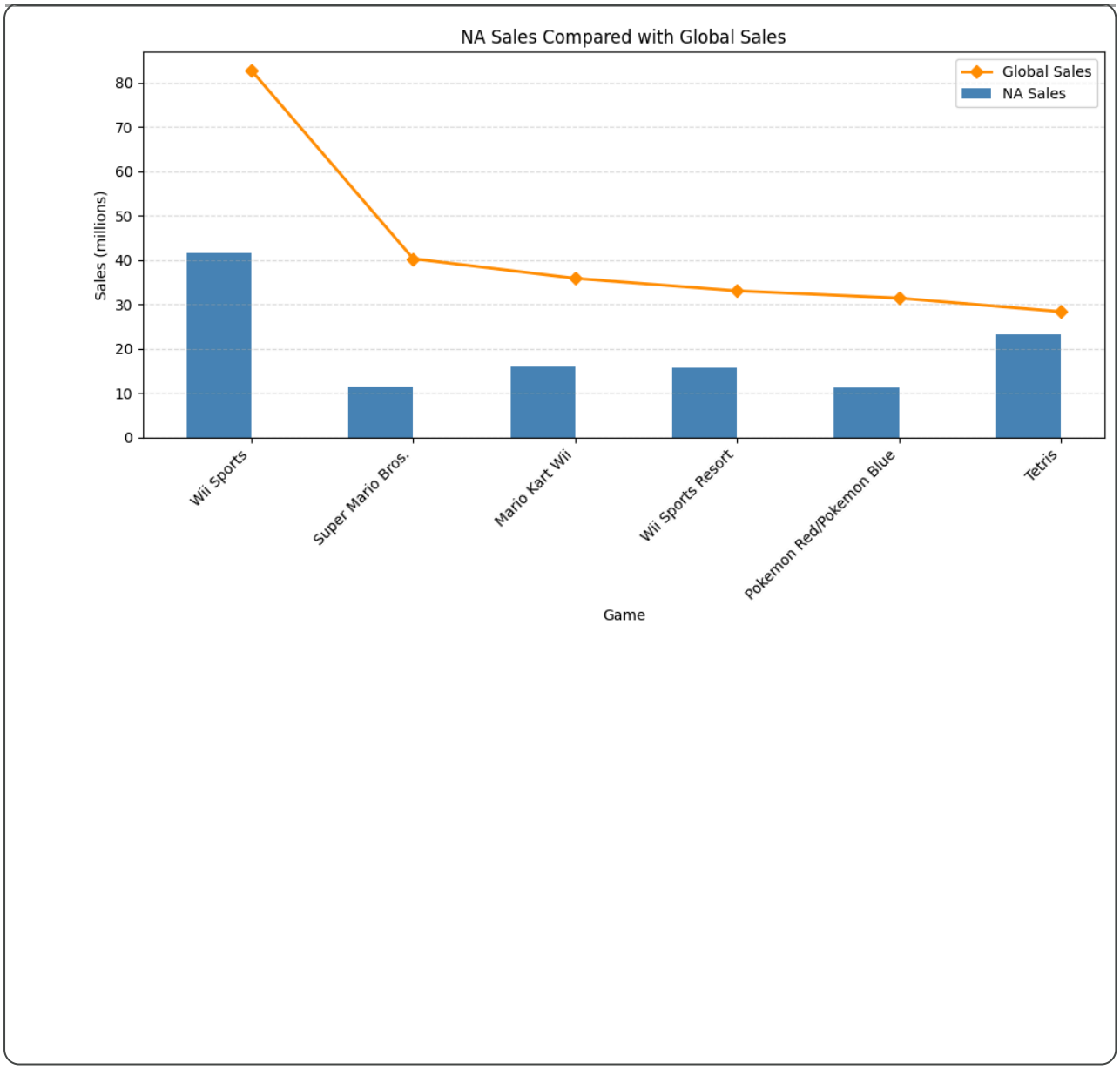
plt.xticks(
    x,
    selected_games["Name"],
    rotation=45,
    ha="right"
)

plt.title("NA Sales Compared with Global Sales")
plt.xlabel("Game")
plt.ylabel("Sales (millions)")

plt.legend()

plt.grid(
    axis="y",
    linestyle="--",
    alpha=0.4
)

plt.tight_layout()
plt.show()
```



Data Story

The dataset contains 19 highly ranked video games released between 1984 and 2013. It includes information about platforms, genres, publishers and sales across North America, Europe, Japan and other regions. Before analysing the data, I checked for missing values and replaced missing numerical sales values with the median.

The results show a large difference between the highest-selling games and the rest. Wii Sports is the strongest game in the dataset, with 82.74 million global sales. It also has particularly strong sales in North America and Europe. Nintendo dominates the dataset, with 16 of the 19 games published by the company, and the Wii is the most common platform.

The scatter plot shows a clear positive relationship between North American and global sales. This suggests that strong performance in North America is generally associated with strong worldwide performance. However, regional preferences are

also visible. Pokémon games perform particularly well in Japan, while several Wii and Nintendo games have strong sales across multiple regions.

Based on these findings, the company should continue targeting the North American market while adapting marketing and game content to regional preferences, particularly in Japan and Europe.